QSVM Track — Week 5

Day-by-day: baselines, diagnostics, and the first significance tests

2 August 2026

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Open (20s). Week 5 for the QSVM track. Seven days, six of them mine.
Three results this week change conclusions already published; the rest close open items.
I will go day by day.

The week at a glance

Day	Deliverable	Outcome
29	EMBER four-model baseline	**Our margins were understated**
30	SOREL-20M close-out	Closed, not deferred
31	Separability + subsample fidelity	Subsamples verified; dataset effect isolated
32	Quantum feature extraction	*Not my scope* (Elizabeth / Ge)
33	Kernel diagnostics	**CIC collapse mechanism identified**
34	Significance tests	**84/84 pairs at** $p < 0.05$
35	Consolidation + paper draft	Backlog empty

Takeaway: three days changed a conclusion. The other three closed something that had been open for weeks.

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Agenda (40s). Seven days. Day 32 is Elizabeth and Ge's — I flag it so nobody reads its absence as a gap on my side.

Bold rows are where a number or a claim moved. Days 30 and 31 are closures.

Day 29 and Day 34 both widen the classical–quantum gap relative to what was reported.

Setup — what makes the comparison fair

CIC-MalMem-2022 — memory dumps, 55 forensic features

- binary: 58 596 rows, balanced
- 15-class: 29 298 malware rows, 1.7× imbalanced

EMBER 2018 — static PE, 2381 features

- binary: 18 014 rows, 88.9% malware
- 15-class: 14 325 rows, 955 per family

Protocol

- quantum kernel is $O(n^2) \Rightarrow$ stratified 1000-row subsample
- subsample + 5 folds *persisted, then replayed* by the classical runner — identical rows, identical folds
- one shared pipeline ending in PCA to n_c
- n_c fixes classical feature count *and* qubit budget at once

Metric is macro-F1, not accuracy. EMBER binary at $n_c = 1$: the QSVM reaches 0.890 accuracy in 3 of 5 folds while benign-class F1 is exactly 0.000 — 88.9% of rows are malware and it labels everything malware.

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Setup (60s). Skip if the audience knows it. One number matters here: n_c. It sets the classical feature count and the qubit budget *simultaneously*. That is what makes this apples-to-apples. The quantum side runs first and writes its rows and folds to disk; the classical side replays them. Nobody picks their own split.

Accuracy example: 0.890 accuracy, benign F1 exactly zero. Hence macro-F1 throughout.

Day 29 — EMBER four-model classical baseline

Every EMBER margin on record compared the QSVM against *SVM alone*. Added random forest, XGBoost, LightGBM:

	Random forest	XGBoost	LightGBM	SVM	QSVM angle	QSVM IQP
Binary						
$n_c = 1$	**0.5582**	0.5451	0.5093	0.5314	0.4532	0.4587
$n_c = 3$	**0.6575**	0.6465	0.6434	0.6306	0.4924	0.5084
$n_c = 6$	**0.6835**	0.6680	0.6593	0.6724	0.5138	0.5137
15-class						
$n_c = 1$	**0.5562**	0.5313	0.5410	0.5294	0.1688	0.1541
$n_c = 3$	0.7194	0.7089	0.7035	**0.7389**	0.3908	0.4797
$n_c = 6$	0.7903	0.7659	0.7709	**0.7930**	0.4232	0.5195

Takeaway: random forest beats SVM on *every* binary cell — the previous baseline was sub-optimal.

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Takeaway: random forest beats SVM on *every* binary cell — the previous baseline was sub-optimal.

Day 29, part 1 (60s). Motivation: we only ever ran SVM on EMBER, so every EMBER margin we published was QSVM-versus-SVM, not QSVM-versus-best.

Point at the bold column: random forest wins all three binary rows.

On 15-class SVM holds up at n_c 3 and 6 — those two margins stand.

Cost: 32 minutes for the whole grid, six foreground runs.

Day 29 — Effect on the reported margins

	$n_c = 1$	$n_c = 3$	$n_c = 6$
gap vs. SVM (as previously reported)	$+0.0727$	$+0.1222$	$+0.1587$
gap vs. *best* classical	$+\mathbf{0.0995}$	$+\mathbf{0.1491}$	$+\mathbf{0.1697}$
understated by	0.0268	0.0269	0.0110

Four of six cells move. All four move the same way.

On 15-class, SVM genuinely *is* the strongest baseline at $n_c \in \{3, 6\}$ — those two margins stand unchanged.

Takeaway: every corrected cell moves against the quantum pipeline. None moves toward it.

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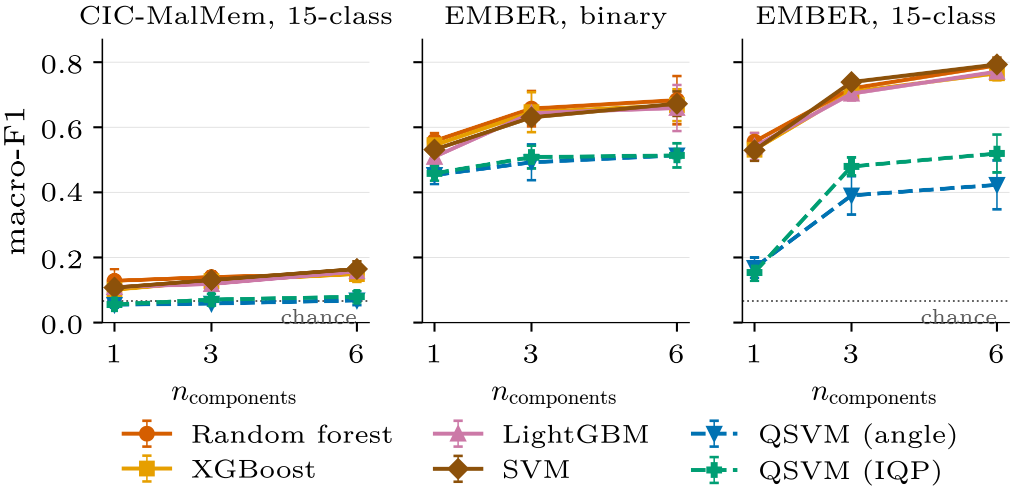
Day 29, part 2 (45s). The gap was under-reported by up to 0.027 macro-F1. Direction matters more than magnitude: the correction is uniformly against quantum. There is no cell where the new baseline helps the QSVM.

If asked "is this material?" — it does not change any conclusion, it makes the existing one stronger.

Day 29 — Capacity scaling — the gap widens



Takeaway: EMBER binary: the QSVM gains 0.055 macro-F1 from 1 to 6 components; the random forest gains 0.125. More qubits are not buying the quantum model the headroom the classical model gets from the same components.

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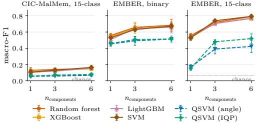

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Day 29, part 3 (40s). One gesture: the two dashed lines stay flat. Solid = classical, dashed = quantum.

The important reading is the *slope*, not the offset. Both improve with more components, but the classical slope is more than twice the quantum one on EMBER binary.

So “more qubits will fix it” is not supported by this curve.

CIC binary is missing from this chart — I explain why on the Day 34 slide.

Day 30 — SOREL-20M — closed, not deferred

Delivered

- labelling scheme designed and validated over 19.7M rows

Never started

- feature-store download
- any sweep, classical or quantum

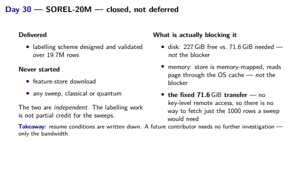
The two are *independent*. The labelling work is not partial credit for the sweeps.

What is actually blocking it

- disk: 227 GiB free vs. 71.6 GiB needed — *not* the blocker
- memory: store is memory-mapped, reads page through the OS cache — *not* the blocker
- **the fixed 71.6** GiB **transfer** — no key-level remote access, so there is no way to fetch just the 1000 rows a sweep would need

Takeaway: resume conditions are written down. A future contributor needs no further investigation — only the bandwidth.

Day 30 — SOREL-20M — closed, not deferred



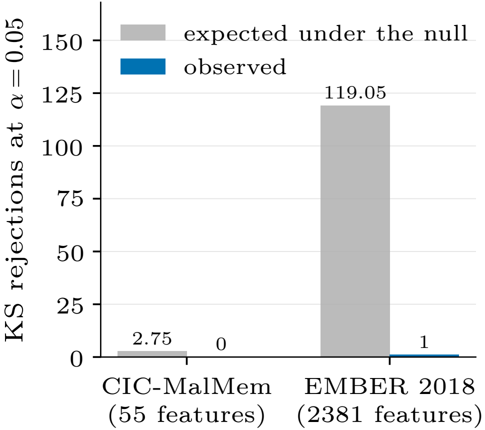
Day 30 (40s). Short slide. Two separate pieces of work; only one was done — the labelling report contains no results.

Zero SOREL rows exist in our experiment log. I checked.

The useful part is the right column: I ruled out disk and memory by measurement, so the next person does not re-investigate. It is purely download time.

Closing rather than deferring because nobody intends to pick it up this cycle.

Day 31 — Are the 1000-row subsamples even faithful?



Specified in Week 2. **Never executed.** Every quantum result in this project has assumed it for four weeks.

A KS rejection count alone is uninterpretable — a *faithful* subsample still rejects $\approx \alpha \cdot n_{\text{features}}$ by chance. So read it against that expectation:

- CIC: **0** of 55 rejected, ≈ 2.75 expected
- EMBER: **1** of 2381 rejected, ≈ 119 expected
- class-proportion drift < 0.001 on both

Takeaway: both subsamples are faithful. Four weeks of results now rest on a measured foundation instead of an assumption.

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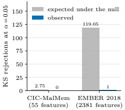

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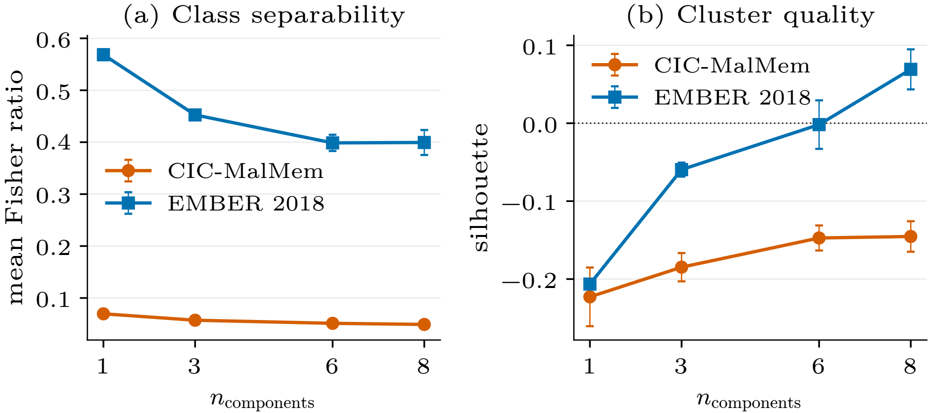
Day 31, part 1 (45s). Assigned in Week 2; not executed until now.
The chart is the whole argument — grey is what chance produces, blue is what we observed.
Both far below chance.

Why it matters: every quantum number in four weeks of reports sits on these 1000 rows.

Day 31 — Feature geometry — a dataset effect



EMBER's families are $\approx 8\times$ more separable by Fisher ratio, up to $50\times$ on the worst-case family. CIC's silhouette is **negative at every n_c**: the average point lies closer to *another* family's centroid than its own. Yet CIC's PCA explains *more* variance (33–81% vs. 8–30%) — its variance sits in directions that do not discriminate family.

Takeaway: this explains why *every* method does badly on CIC 15-class. It cannot explain the QSVM-specific failure.

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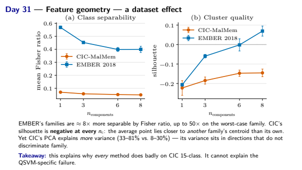

Day 31, part 2 (50s). Set up the Day 33 punchline here. Both metrics are scale-invariant, so the CIC-vs-EMBER comparison is legitimate despite very different dimensionality.

The counter-intuitive bit is worth pausing on: CIC's PCA captures *more* variance and separates *less*. Its variance tracks workload, not family.

Then the pivot: classical SVM gets 0.108–0.165 from these same features where the QSVM gets 0.056–0.079. So hard features cannot be the whole story.

That question is Day 33.

Day 32 — Quantum feature extraction — not my scope

Day 32 (quantum feature extraction, C0-V0-H2) belongs to **Elizabeth and Ge**.

Listed here so its absence is not read as a gap on the QSVM side.

Also not mine, and listed for the same reason: VQC (not planned by this contributor) · CTGAN run (blocked on Team B) · evaluation-protocol finalisation (blocked on Team A)

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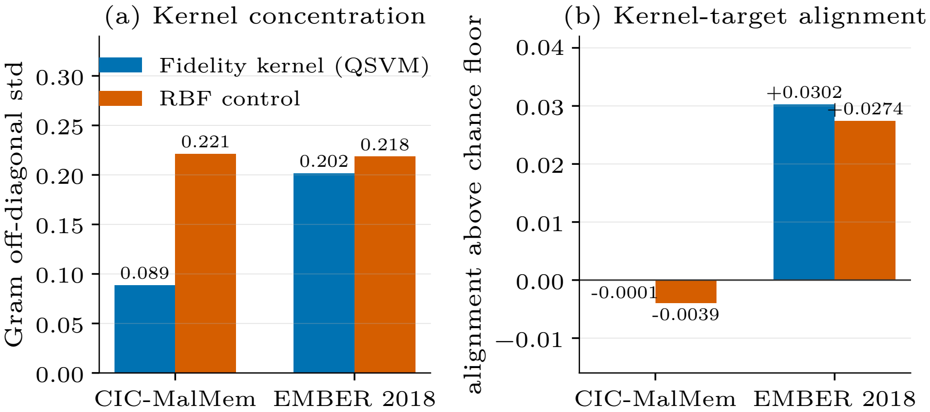
Day 32 (20s). Fast slide. Present so the day sequence is complete.
If someone asks about the greyed line: CTGAN is ready on my side — the CLI takes any parquet — it just needs the augmented dataset from Team B.

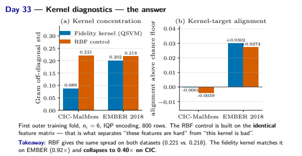
Day 33 — Kernel diagnostics — the answer



First outer training fold, $n_c = 6$, IQP encoding, 800 rows. The RBF control is built on the **identical** feature matrix — that is what separates "these features are hard" from "this kernel is bad".

Takeaway: RBF gives the same spread on both datasets (0.221 vs. 0.218). The fidelity kernel matches it on EMBER (0.92×) and **collapses to 0.40× on CIC**.

Day 33 — Kernel diagnostics — the answer



Day 33 (70s). Densest slide — take it slowly. The control is the whole design: same features, two kernels.

Panel (a), read the orange bars first — RBF is flat across datasets. So CIC's features do *not* make a classical kernel concentrate.

Then the blue: matches RBF on EMBER, collapses on CIC. That is a property of the *kernel*.

Panel (b): alignment must be read against its floor. A constant kernel scores $1/\sqrt{15}$ on 15 balanced classes. On CIC the fidelity kernel is *below* that floor — indistinguishable from an all-ones matrix.

Cost note if asked: 6 minutes, two Gram builds. The sweep-wide concentration numbers were already logged, so they were free.

Day 33 — What this supports — and what it does not

Supported

The failure on CIC is a property of **the kernel**, not of the dataset. The Gram approaches a scaled identity — every pair looks equally similar — and an SVM given that has nothing to separate on. That is what a macro-F1 pinned to chance looks like from the inside.

Not supported

That tuning `bandwidth` would repair it. Bandwidth is the identified mechanism and has never been tuned on CIC beyond its default — but no sweep was run. A hypothesis with a named lever, not a measured fix.

Unresolved: alignment vs. SVM performance

Alignment says neither kernel carries label information on CIC, yet classical SVM reaches 0.16 there. Alignment is a global unweighted average over all pairs; an SVM needs no such thing. It ranks dataset difficulty well and predicts achievable SVM performance poorly.

Day 33 — What this supports — and what it does not

Day 33, part 2 (50s). CIC is not fixed — the mechanism is named, the lever is untested. Third block: alignment and SVM performance disagree on CIC. Alignment averages over all pairs; concentration describes the matrix the SVM receives, which is why the two can diverge.

Day 33 — What this supports — and what it does not

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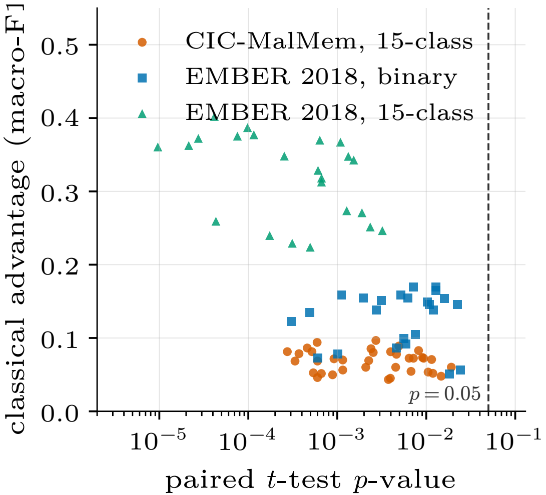
Day 34 — Significance tests — specified in Week 1, never run



The infrastructure was built and unit-tested in Week 1 and **never executed**.

Four weeks of "QSVM loses to classical" rested on differences in mean macro-F1, with no test that any gap exceeded fold-to-fold noise.

All 84 testable pairs favour the classical model, and all 84 reach $p < 0.05$ on the paired t-test.

Worst p-value across every pair: **0.0241**.

Takeaway: the headline claim now has evidence behind it, not an unexamined mean difference.

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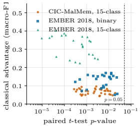

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Day 34, part 1 (45s). The claim ran for four weeks untested.
Every point is left of the dashed line and above zero. 84 out of 84.
Worst p-value is 0.024 — so this is not marginal anywhere.

Next slide is the caveat slide; do not stop here or it sounds too clean.

Day 34 — Limits of the test

Wilcoxon returned 0.0625 for all 84 pairs — that is not a null result

With 5 paired samples, 0.0625 is the *smallest attainable* two-sided p-value. The test cannot reach 0.05 at this fold count, whatever the effect size. All 84 hitting the floor means every pair achieved the most extreme rank configuration available. *Fix:* 6 folds puts 0.03125 in reach — not a different test.

All CIC binary comparisons are untestable

Every CIC binary run — classical **and** quantum — predates parent/child run tracking and carries no recoverable sweep boundary. Raw records show the QSVM at 0.92–1.00, consistent with the earlier "ties on binary" reading, but they resolve into two sweeps per n_{c} with no identifier separating them. *We did not reconstruct a grouping* — that would invent a boundary the data does not record.

Takeaway: symmetric across frameworks — a logging gap, not a quantum result.

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Day 34, part 2 (55s). Expect questions here. 0.0625 reads as failure but is the floor at $n = 5$; “Wilcoxon found nothing” would misstate it.

CIC binary is the only configuration where quantum was competitive, and the one we cannot test.

If asked why not group by timestamp: the gaps are not a recorded boundary, so any grouping would be inferred rather than measured.

Day 35 — Where the week leaves us

- Classical models beat the QSVM on **every** testable configuration — now with $p < 0.05$ on all 84, not on mean differences alone.
- The EMBER margins were **understated**, not overstated. Every correction this week went against the quantum pipeline.
- CIC binary — the one configuration where quantum was ever competitive — cannot be tested from what was logged.

The transferable result

The fidelity kernel does not fail on CIC because CIC's features are hard — a classical RBF kernel on identical features does not concentrate. It fails because **the fidelity kernel collapses to $\mathbf{0.40\times}$ the classical kernel's spread**. That is a statement about the method, not the dataset.

Next: tune `bandwidth` on CIC — the named mechanism, still untested. Move to 6 folds so a non-parametric test becomes usable.

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Day 35, close (60s). Three bullets, then one idea.

The last block is the transferable finding: it is about the *method*, not either dataset, so it carries to other data.

Backlog is empty on the QSVM side — 144 tests passing, reports and paper sections written.

Two concrete next steps, both small. Bandwidth on CIC is the obvious follow-up because we named the mechanism and did not test the lever.

Stop here. Take questions.